

Logan Boswell

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EDUCATION

Northwestern University
Master of Science in Robotics

Evanston, IL
December 2025

Georgia Institute of Technology
Bachelor of Science in Mechanical Engineering (Highest Honors)
Concentration in Automation and Robotics

Atlanta, GA
May 2024

EXPERIENCE

Southwest Research Institute
Robotics Engineer

San Antonio, TX
Feb 2026 - Present

- Supporting hardware design and software development efforts to build advanced robotics solutions

GrayMatter Robotics
Robotics Engineering Intern

Carson, CA
Jun - Sep 2025

- Automated surface finishing for acrylic jet canopies through wet sanding and polishing with Fanuc M-710 robots
- Extended Dockerized ROS/ROS 2 packages using C++ and Python to support high-resolution camera inspection
- Optimized finishing process parameters to remove scratches and defects while minimizing distortions

Georgia Tech Research Institute
Student Assistant - Part Time

Smyrna, GA
Jan - May 2024

- Collaborated with electrical engineers to design housings and mounts for RF systems using SolidWorks
- Operated mills, water jets, and sheet metal folders to fabricate components for electro-mechanical systems

Chick-fil-A Corporate Support Center
Equipment and Systems Engineering Co-Op

Atlanta, GA
Jan 2022 - Aug 2023

- Developed code libraries using Python, C++, and Git for an automated frying mechatronic system
- Modeled an electric heater in SolidWorks to meet strict size and watt density specifications
- Created a user-friendly GUI with Python and PyQt5 to streamline operation of an automated frying system

PROJECTS

Cable-Driven Parallel Robot for Submerged Applications

Apr - Jun, Sep - Dec 2025

- Designed a planar cable-driven parallel robot in Onshape capable of operating while submerged in water
- Developed C++ software on a Teensy 4.1 to control ODrive S1 drivers via CAN bus and handle serial commands
- Implemented a custom force-based PID + feedforward control system for precise end-effector movements

Ping Pong Robot from Scratch

Jan - Mar 2025

- Designed and built a 3-wheeled omnidirectional robot using Onshape and rapid prototyping methods
- Developed a ROS 2 package in C++ for operating the robot and integrating kinematics with computer vision
- Utilized OpenCV, AprilTags, and a RealSense camera for ping pong ball object detection and tracking

7-DOF Pool-Playing Robot

Nov - Dec 2024

- Collaborated with a team of 5 to develop a Python ROS 2 package for a Franka Panda arm to play a game of pool
- Wrote a Python ROS 2 API wrapper to plan and execute trajectories using MoveIt

Propeller-Driven Balance Beam Control System

Oct - Dec 2023

- Guided a team of 4 to design two PID controllers to balance a ball in the center of a beam and reject disturbances
- Wrote C++ code to incorporate PID controllers for stabilizing an inherently unstable physical system

SKILLS

Robotics: ROS/ROS 2, Control Systems, Embedded Systems, MoveIt, Nav2, OpenCV, RVIZ, Gazebo

Software: C++, Python, C, Embedded C, MATLAB/Simulink, LabVIEW, Linux, Git, CMake, Docker

Hardware: SolidWorks/NX/Onshape, EagleCAD, Machining, 3D Printing, Soldering, Microcontrollers, FEA